

Session 2020-21 Question Paper

Dept. of Computer and Communication Engineering

Faculty of Computer Science and Engineering

Patuakhali Science and Technology University

Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: 3 Semester: II Session: 2020-21
Course Code CCE-323 **Course Title** Optical Fiber Communication **July-December 2023** **Credit: 03**
Time: 03 Hr **Marks: 70**

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1	[A]	i.	A long single-mode optical fiber has an attenuation of 0.5 dB km^{-1} when operating at a wavelength of $1.3 \mu\text{m}$. The fiber core diameter is $6 \mu\text{m}$ and the laser source bandwidth is 600 MHz. Compare the threshold optical powers for stimulated Brillouin and Raman scattering within the fiber at the wavelength specified.	4
		ii.	Describe different advantages of optical fiber communication.	3
	[B]	i.	A 10 km length of fiber with a mean optical power launch of $120 \mu\text{W}$ will have a mean optical power of $4 \mu\text{W}$ at the fiber output. Assess: a) If no connectors or splices are present, the total signal attenuation or loss in decibels via the fiber; b) the fiber's signal attenuation per kilometer; c) the total signal attenuation for a 20 km optical link made of the same fiber with 2 km-separated splices that each provide a 2 dB attenuation; d) the numerical input/output power ratio in (c).	4
		ii.	What is optical fiber? Show its typical configuration with necessary diagram.	3
2	[A]		What are the sources of fiber optic losses in silica? Provide a comprehensive explanation of the intrinsic absorption loss mechanisms, including relevant examples.	7
	[B]		Two step index fibers exhibit the following parameters: a) a multimode fiber with a core refractive index of 1.500, a relative refractive index difference of 3% and an operating wavelength of $0.82 \mu\text{m}$; b) a $8 \mu\text{m}$ core diameter single-mode fiber with a core refractive index the same as (a), a relative refractive index difference of 0.3% and an operating wavelength of $1.55 \mu\text{m}$. Estimate the critical radius of curvature at which large bending losses occur in both cases.	7
3	[A]		What sets an optical fiber communication system apart from a conventional	7

		communication system? Illustrate the optical fiber communication system using an appropriate block diagram.	
	[B]	Silica has an estimated fictive temperature of 1400 K with an isothermal compressibility of $7 \times 10^{-11} \text{ m}^2 \text{N}^{-1}$. The refractive index and the photo elastic coefficient for silica are 1.46 and 0.286 respectively. Determine the theoretical attenuation in decibels per kilometer due to the fundamental Rayleigh scattering in silica at optical wavelengths of 0.63, 1.00 and 1.30 μm . Boltzmann's constant is $1.381 \times 10^{-23} \text{ JK}^{-1}$.	7
4	[A]	What are the direct and indirect Band Gap Semiconductors?	3
	[B]	Explain the working principle of Expanded Beam Connectors used in optical fiber communication. Discuss their advantages and disadvantages compared to physical contact fiber connectors.	4
	[C]	Discuss the mechanism of optical feedback to provide oscillation and hence amplification within the laser. Indicate how this provides a distinctive spectral output from the device.	4
	[D]	The radiative and non-radiative recombination lifetimes of the minority carriers in the active region of a double-heterojunction LED are 60 ns and 100 ns respectively. Determine the total carrier recombination lifetime and the power internally generated within the device when the peak emission wavelength is 0.87 μm at a drive current of 40 mA.	3
5	[A]	Describe, two common techniques used for the mechanical splicing of optical fibers with the suitable diagrams.	4
	[B]	Explain the operation of a substrate-entry InGaAs p-i-n photodiode by: i. Drawing and labeling its layer structure. ii. Sketching the energy band diagram and discussing the effects of the heterojunction on charge trapping and carrier dynamics.	4
	[C]	Two multimode step index fibers have numerical apertures of 0.2 and 0.4, respectively, and both have the same core refractive index of 1.48. Estimate the insertion loss at a joint in each fiber caused by a 5° angular misalignment of the fiber core axes. It may be assumed that the medium between the fibers is air.	3
	[D]	What is the Fermi-Dirac distribution, and how does it describe the probability of electron occupancy in energy states of a semiconductor?	3
6	[A]	With the aid of simple sketches outline the major categories of multiport optical fiber coupler. Describe two common methods used in the fabrication of three- and four-port fiber couplers.	5
	[B]	A photodiode has a quantum efficiency of 65% when photons of energy $1.5 \times 10^{-19} \text{ J}$ are incident upon it. (i) At what wavelength is the photodiode operating? (ii) Calculate the incident optical power required to obtain a photocurrent of 2.5 μA when the photodiode is operating as described above.	3
	[C]	What is the main difference between LED and Laser?	2
	[D]	How does the double-heterojunction structure enhance the performance of an LED? Support your answer with a labeled diagram of the layer structure under forward bias and the corresponding energy band diagram.	4

1	[A]	What distinguishes an optical fiber communication system from a normal communication system? Using the relevant block diagram, describe the optical fiber communication system.	7
	[B]	An 8 km length of fiber with a mean optical power launch of 120 μ W will have a mean optical power of 3 μ W at the fiber output. Assess: i. If no connectors or splices are present, the total signal attenuation or loss in decibels via the fiber; ii. The fiber's signal attenuation per kilometer. iii. the total signal attenuation for a 20 km optical link made of the same fiber with 2 km-separated splices that each provide a 2 dB attenuation; iv. The numerical input/output power ratio in (c).	7
2	[A]	What are the origins of fiber optic losses in silica? Give a thorough explanation of the intrinsic absorption loss mechanisms along with any required examples.	7
	[B]	The parameters of two step index fibers are as follows: i. a multimode fiber with an operational wavelength of 0.82 μ m, a relative refractive index difference of 3%, and a core refractive index of 1.500; ii. a single-mode fiber with an 8 μ m core diameter, an operating wavelength of 1.55 μ m, a relative refractive index difference of 0.3%, and the same core refractive index as (a). Estimate the critical radius of curvature at which large bending losses occur in both cases.	7
3	[A]	i. What is optical fiber? Show its typical configuration with necessary diagram. ii. Describe different advantages of optical fiber communication.	7

	[B]	Silica has an estimated fictive temperature of 1400 K with an isothermal compressibility of $7 \times 10^{-11} \text{m}^2 \text{N}^{-1}$. The refractive index and the photo elastic coefficient for silica are 1.46 and 0.286 respectively. Determine the theoretical attenuation in decibels per kilometer due to the fundamental Rayleigh scattering in silica at optical wavelengths of 0.63, 1.00 and 1.30 μm . Boltzmann's constant is $1.381 \times 10^{-23} \text{JK}^{-1}$.	7
4	[A]	What is meant by the fusion splicing of optical fibers? Explain the basic arc fusion method with pre-fusion method for accurately splicing optical fibers. Draw Electric arc fusion splicing diagram.	6
	[B]	Briefly describe the major reasons for the cabling of optical fibers which are to be placed in a field environment. State the functions of the optical fiber cable.	3
	[C]	Draw and explain a degenerate p-n junction where the position of the Fermi level and the electron occupation (shading) with applied bias and with no applied bias will be clearly depicted.	5
5	[A]	Define the two processes by which light can be emitted from an atom. Discuss the requirement for population inversion in order that stimulated emission may dominate over spontaneous emission. Illustrate your answer with an energy level diagram of a common non-semiconductor laser.	4
	[B]	Explain mechanical splicing techniques that utilize grooves to secure optical fibers during the jointing process, and how these grooves contribute to the overall stability and alignment of the splice?	5
	[C]	What is the main difference between LED and Laser? Draw and distinguish among different type of misalignment in fiber joint which are source of loss at a fiber-fiber connection.	5
6	[A]	Describe, with the aid of suitable diagrams, the major strategies and structures utilized in the fabrication of single-frequency injection lasers (The double-heterojunction injection laser). Indicate the reasons for the great interest in such devices.	5
	[B]	A four-port multimode fiber FBT coupler has 60 μW optical power launched into port 1. The measured output powers at ports 2, 3 and 4 are 0.004, 26.0 and 27.5 μW respectively. Determine the excess loss, the insertion losses between the input and output ports, the crosstalk and the split ratio for the device.	3
	[C]	i. Draw schematic illustration of the vapor-phase deposition techniques used in the preparation of low-loss optical fibers. ii. Briefly explain outside vapor-phase oxidation process.	6

Session 2018-2019 Question Paper

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Course Code	Course Title	Credit: 03
CCE-323	Optical Fiber Communication	Time: 03 Hr
		Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1	[A]	Define following terms with respect to optical laws – a) Critical angle and acceptance angle b) Refractive index	4
	[B]	Discuss the requirement for population inversion in order that stimulated emission may dominate over spontaneous emission. Illustrate your answer with an energy level diagram of a common non-semiconductor laser.	4
	[C]	"Longitudinal misalignment gives significantly minor losses per unit displacement than the Lateral misalignment"- True or false, justify your statement.	2
	[D]	Draw and explain p-n junction with strong forward bias hence stimulated emission is obtained in this region.	4
2	[A]	Explain the different types of rays in fiber optic.	4
	[B]	Briefly discuss, with the aid of suitable diagrams, the following concepts in optical fiber transmission: a) Modes in a planar guide b) Goos-Haenchen shift;	6
	[C]	A step index fiber has a core refractive index of 1.5 and a core diameter of 50 μm . The fiber is jointed with a lateral misalignment between the core axes of 5 μm . Estimate the insertion loss at the joint due to the lateral misalignment assuming a uniform distribution of power between all guided modes when: a) there is a small air gap at the joint; b) the joint is considered index matched.	4
3	[A]	Describe the three types of fiber misalignment which may contribute to insertion loss at an optical fiber joint with appropriate figure.	6
	[B]	Derive the expression for electromagnetic mode theory for optical propagation. i. Electromagnetic waves ii. Phase and group velocity	4
	[C]	Discuss the mechanism of optical feedback to provide oscillation and hence amplification within the laser. Indicate how this provides a distinctive spectral output from the device.	4
4	[A]	Write short note on ST connector, SC connector and FC connector.	5
	[B]	Describe the operational principle of optical coupler using 2 X 2 Coupler.	3
	[C]	Write the working principle of wavelength converter.	6
5	[A]	Describe Optical isolators and Optical circulators.	6
	[B]	What is an Optical Switch? Describe the working procedure of optical switch.	6

		a) a multimode fiber with a core refractive index of 1.500, a relative refractive index difference of 3% and an operating wavelength of 0.82 μm; b) an 8 μm core diameter single-mode fiber with a core refractive index the same as (a), a relative refractive index difference of 0.3% and an operating wavelength of 1.55 μm. Estimate the critical radius of curvature at which large bending losses occur in both cases.	
3	[A.]	i. What are the differences between the overall communication system and the optical fiber communication system? ii. Explain the fiber optic communication system with a suitable block diagram.	7
	[B.]	Silica has an estimated fictive temperature of 1400 K with an isothermal compressibility of $7 \times 10^{-11} \text{m}^2 \text{N}^{-1}$. The refractive index and the photo-elastic coefficient for silica are 1.46 and 0.286 respectively. Determine the theoretical attenuation in decibels per kilometer due to the fundamental Rayleigh scattering in silica at optical wavelengths of 0.63, 1.00 and 1.30 μm. Boltzmann's constant is $1.381 \times 10^{-23} \text{JK}^{-1}$.	7
4	[A.]	Discuss optical fiber cable design with regard to: a) fiber buffering; b) cable strength and structural members	
	[B.]	How does single-frequency device operation of the distributed feedback (DFB) laser happens in comparison with the Fabry-Pérot laser?	4
	[C.]	Describe two common methods used in the fabrication of three- and four-port fiber couplers.	4
5	[A.]	Discuss the mechanism of optical feedback to provide oscillation and hence amplification within the laser. Indicate how this provides a distinctive spectral output from the device.	4
	[B.]	Describe what is meant by the fusion splicing of optical fibers. Discuss the advantages and drawbacks of this jointing technique.	4
	[C.]	A four-port multimode fiber FBT coupler has 60 μW optical power launched into port 1. The measured output powers at ports 2, 3 and 4 are 0.004, 26.0 and 27.5 μW respectively. Determine the excess loss, the insertion losses between the input and output ports, the crosstalk and the split ratio for the device.	3
	[D.]	What are the direct and indirect Band Gap Semiconductors?	3
6	[A.]	Draw and explain p-n junction with strong forward bias hence stimulated emission is obtained in this region.	4
	[B.]	Briefly describe in general terms liquid-phase techniques for the preparation of multicomponent glasses for optical fibers.	7
	[C.]	An optical fiber has a core refractive index of 1.5. Two lengths of the fiber with smooth and perpendicular (to the core axes) end faces are butted together. Assuming the fiber axes are perfectly aligned, calculate the optical loss in decibels at the joint (due to Fresnel reflection) when there is a small air gap between the fiber end faces.	3